TASK -1 — GROUPING & AGGREGATION

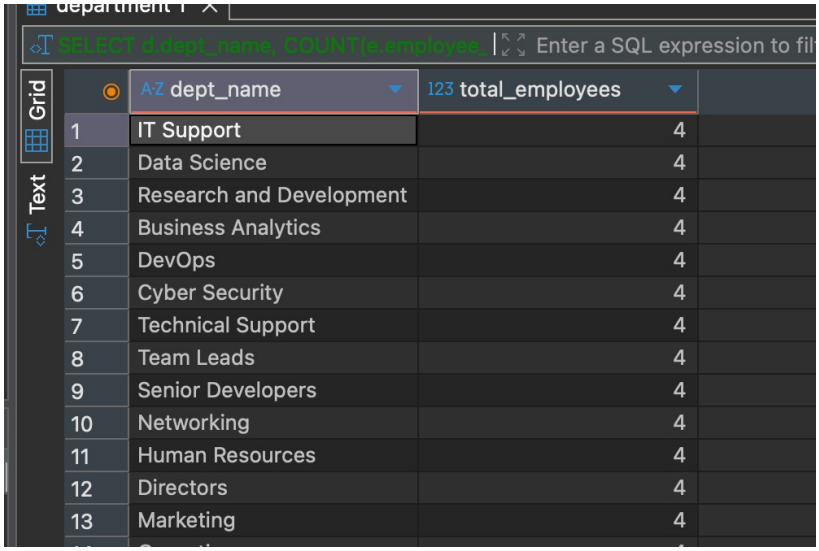
Task A — Basic Counts

Q1) Find the number of employees per department.

Query:

```
SELECT d.dept_name,
       COUNT(e.employee_id) AS total_employees
FROM employee e
JOIN department d
ON e.dept_id = d.dept_id
GROUP BY d.dept_name;
```

Explanation:
This query calculates the total number of employees in each department. It joins the employee and department tables using the dept_id column and groups the results by department name.



	dept_name	total_employees
1	IT Support	4
2	Data Science	4
3	Research and Development	4
4	Business Analytics	4
5	DevOps	4
6	Cyber Security	4
7	Technical Support	4
8	Team Leads	4
9	Senior Developers	4
10	Networking	4
11	Human Resources	4
12	Directors	4
13	Marketing	4

Task B — Condition on Groups using HAVING

Q2) List departments having more than 2 employees.

Query:

```
SELECT d.dept_name,
       COUNT(e.employee_id) AS total_employees
FROM employee e
JOIN department d
ON e.dept_id = d.dept_id
GROUP BY d.dept_name
HAVING COUNT(e.employee_id) > 2;
```

Explanation:
This query identifies departments with more than two employees. It groups employees by department name and uses the HAVING clause to filter groups based on the employee count.

department 1

SELECT d.dept_name, COUNT(e.employee_ | Enter a SQL expression to fil

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	dept_name	total_employees
1	IT Support	4
2	Data Science	4
3	Research and Development	4
4	Business Analytics	4
5	DevOps	4
6	Cyber Security	4
7	Technical Support	4
8	Team Leads	4
9	Senior Developers	4
10	Networking	4
11	Human Resources	4
12	Directors	4
13	Marketing	4

Task C — Statistics per Group

Q3) Find the average team size per project.

Query:

```
SELECT p.project_name,
      COUNT(e.employee_id) * 1.0 / COUNT(DISTINCT t.team_id)
         AS avg_team_size
FROM project p
JOIN team t
ON p.project_id = t.project_id
JOIN employee e
ON t.team_id = e.team_id
GROUP BY p.project_name;
```

Explanation:
This query calculates the average team size for each project. It counts the total number of employees and divides it by the number of distinct teams per project, grouping the results by project name.

```
SELECT p.project_name, COUNT(e.employe
```

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	project_name	avg_team_size
1	AI Chatbot	4
2	Analytics Dashboard	4
3	Banking App	4
4	CRM Upgrade	4
5	Cloud Migration	4
6	Cybersecurity Audit	4
7	Data Warehouse	4
8	Digital Marketing	4
9	Document Management	4
10	E-commerce Portal	4
11	ERP Deployment	4